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Lab 1: Zabbix Installation

Lab Workbook

Part 1: Install Ubuntu Server (If you have a previous Ubuntu server running, use that)

Part 2: Install Zabbix

Lab Objectives

Upon completion of this lab, you will be able to perform the following:

- Zabbix Installation

Lab Materials

- Please name your Lab workbook titled "Lab 1 - <name> <Student ID>".
- Install Ubuntu server
- Install Zabbix Server

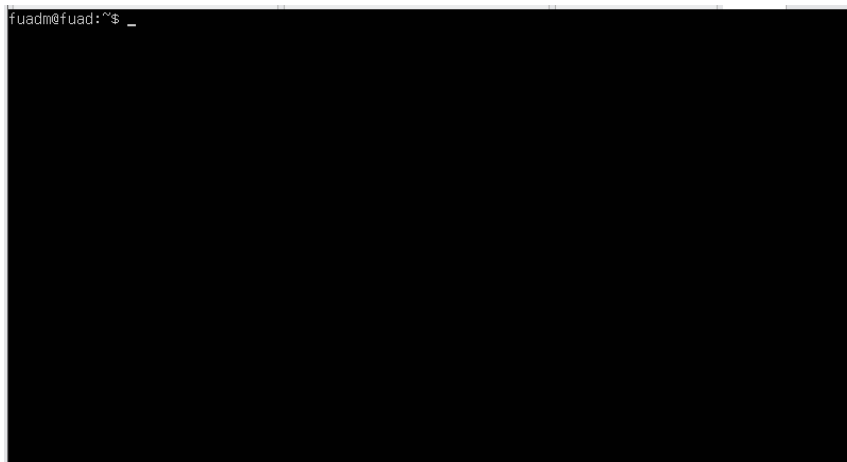
Lab Requirements

- VMware workstation 16 pro or higher Or Virtualbox
- Ubuntu server

Lab 1 Week 2 – Zabbix Server Installation

Part 1: Ubuntu server Installation **(Please keep in mind that if you have a previous Ubuntu server installed on VMware or Virtualbox, you can use it; I do not mean the Ubuntu with a graphical user interface; I mean the server with only command line access.)**

1. Download Ubuntu Server 20.04 from here: <https://ubuntu.com/download/server> (choose “manual server installation” to get the download file)
2. Create a new Virtual Machine (VM) using the Ubuntu Server 20.04 iso from the link above
 - a. For the username setup, **use your first name and last name initials e.g., fuadm**
3. Once the server is installed, login with your created username just as seen below.

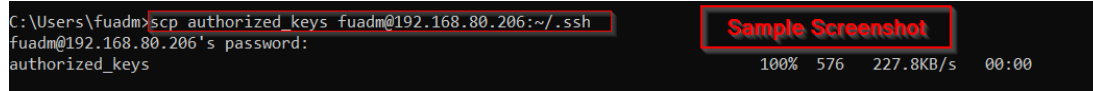


Install SSH on the Ubuntu server using the following instructions (Ignore this if your Ubuntu server has SSH installed).

4. Use the command `ip a` to confirm your server’s IP address or run the command `sudo apt install net-tools` to be able to use `ifconfig` command.
 - a. Make sure that you can ping the IP from your local Windows computer
5. On your local Windows computer **(NOT your ubuntu machine)** create an ssh keypair: open command prompt and type: `ssh-keygen`
6. Once the keypair is created, copy the public key to a safe directory with the command:
`copy C:\Users\yourUserName\.ssh\id_rsa.pub C:\Users\yourUserName\authorized_keys`
7. Go back to your Ubuntu server
8. Start the SSH service using the commands:
`sudo apt install openssh-server`
`sudo ufw allow ssh`
9. switch to the ssh directory with the command: `cd /home/[YourUsername]/.ssh`

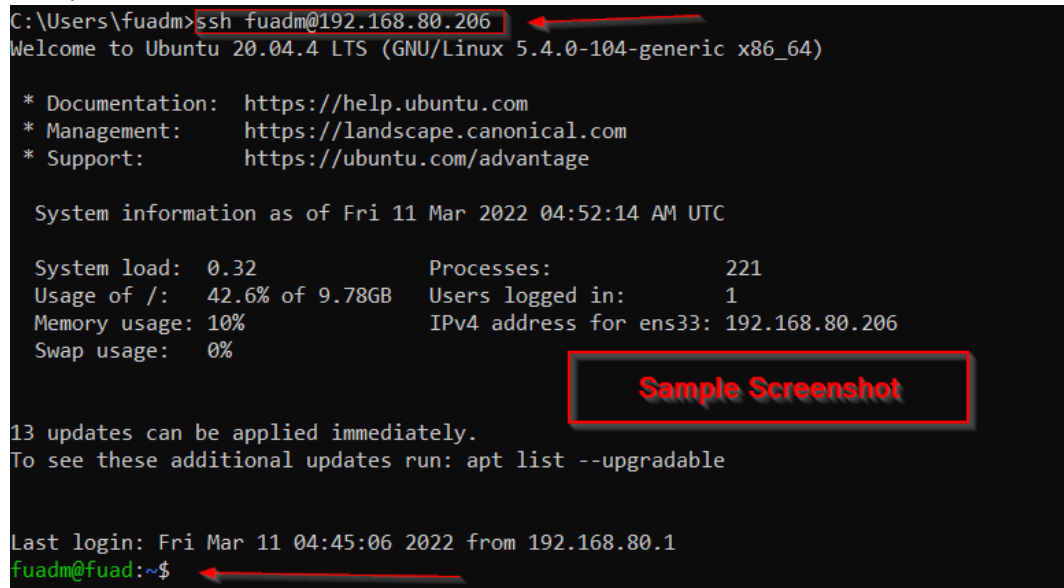
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10. Go back to your Windows host computer
11. Open command prompt and use the command: `cd C:\Users\yourUserName\`
12. Copy your public ssh key to the ubuntu server using the command:
13. `scp authorized_keys username@UbutnuServerIP:~/.ssh`



14. Go back to your Ubuntu server
15. Adjust the permission settings for the ssh share and public ssh key file:
`chmod 700 ~/.ssh`
`chmod 600 ~/.ssh/authorized_keys`
16. Go back to your Windows host computer
17. Log into your Ubuntu machine from the Windows command prompt with the command:
`ssh username@UbuntuServerIP`
- b. You should not be prompted for a password

i. Example:



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Part 2: Installation of the Zabbix server

1. It is recommended to update your VM before installation to ensure all dependencies are updated. Then install the mysql server.

Run the system update of all the packages using the command below

```
$ sudo apt update
```

2. Then install the mysql-server package using the following command:

```
$ sudo apt install mysql-server
```

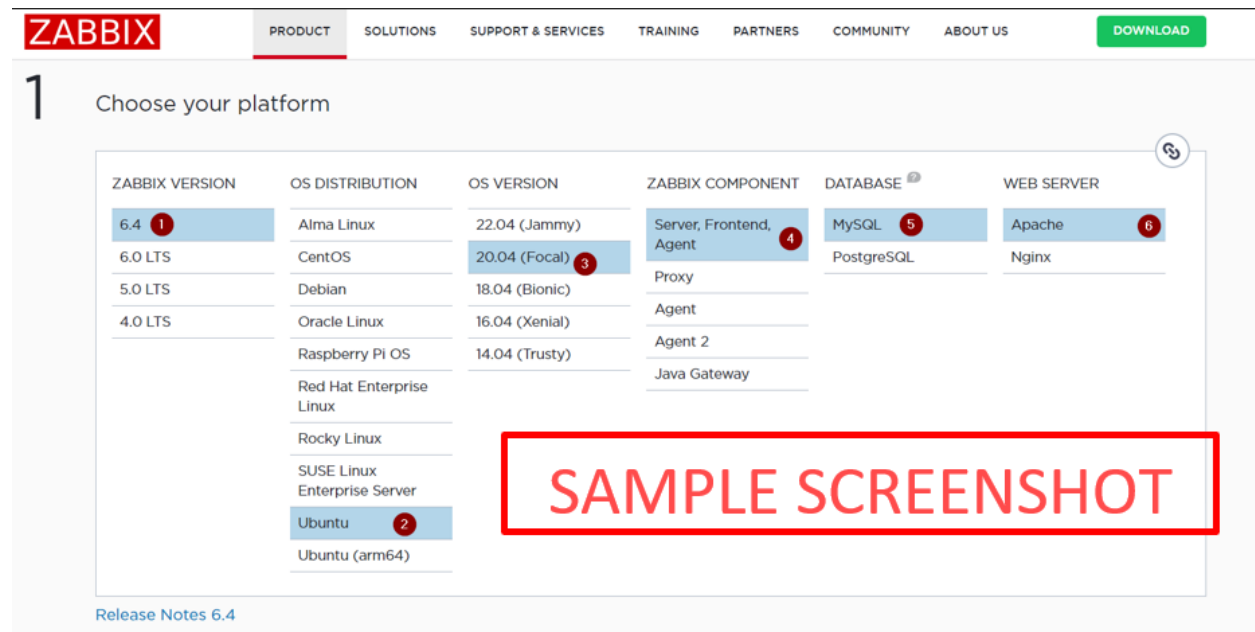
3. Ensure that the server is running using the systemctl start command:

```
$ sudo systemctl start mysql.service
```

4. Go to the link to see the commands for installing Zabbix

https://www.zabbix.com/download?zabbix=6.4&os_distribution=oracle_linux&os_version=9&components=server_frontend_agent&db=mysql&ws=apache

Select the following combinations as seen in the screenshot below.



5. Scroll down the page to see the installation instructions. This shows all the steps needed to be taken to install our Zabbix server. See screenshot below

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2 Install and configure Zabbix for your platform

a. Install Zabbix repository
[Documentation](#)

```
# wget https://repo.zabbix.com/zabbix/6.4/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.4-1+ubuntu20.04_all.deb
# dpkg -i zabbix-release_6.4-1+ubuntu20.04_all.deb
# apt update
```

b. Install Zabbix server, frontend, agent

```
# apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent
```

c. Create initial database
[Documentation](#)

Make sure you have database server up and running.

Run the following on your database host.

```
# mysql -uroot -p
password
mysql> create database zabbix character set utf8mb4 collate utf8mb4_bin;
mysql> create user zabbix@localhost identified by 'password';
mysql> grant all privileges on zabbix.* to zabbix@localhost;
```

- SSH into the Ubuntu server to make it easier to copy and paste your commands. Using the command


```
> ssh username@UbuntuServerIP
> ssh fuadm@192.168.108.131
```

Proceed with installing the zabbix repository using the command in the section above.

```
# wget https://repo.zabbix.com/zabbix/6.4/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.4-1+ubuntu20.04_all.deb
# sudo dpkg -i zabbix-release_6.4-1+ubuntu20.04_all.deb
# sudo apt update
```

- Run the system update once again using the below command.

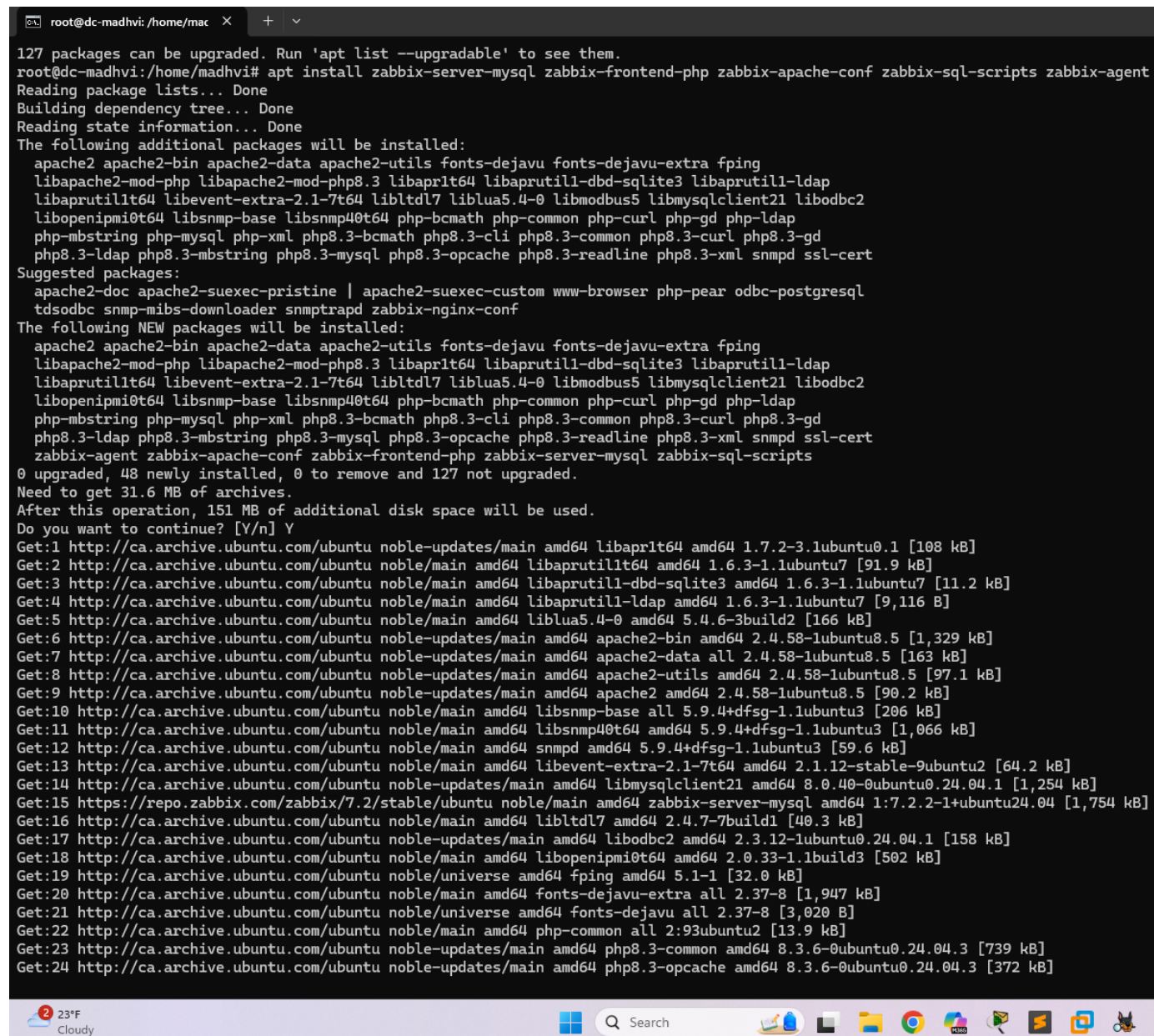

```
$ sudo apt update
```

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8. Next step would be to Install Zabbix server, frontend, agent to the Ubuntu server you installed.

```
$ sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent -y
```

[Attach Your Screenshot]



```
root@dc-madhvi: /home/mac X + v
127 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@dc-madhvi: /home/madhvi# apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils fonts-dejavu fonts-dejavu-extra fping
  libapache2-mod-php libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64 libevent-extra-2.1-7t64 libltdl7 liblua5.4-0 libmodbus5 libmysqlclient21 libodbc2
  libopenipmi0t64 libsnmp-base libsnmp40t64 php-bcmath php-common php-curl php-gd php-ldap
  php-mbstring php-mysql php-xml php8.3-bcmath php8.3-cli php8.3-common php8.3-curl php8.3-gd
  php8.3-ldap php8.3-mbstring php8.3-mysql php8.3-opcache php8.3-readline php8.3-xml snmpd ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser php-pear odbc-postgresql
  tdsodbc snmp-mibs-downloader snmptrapd zabbix-nginx-conf
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils fonts-dejavu fonts-dejavu-extra fping
  libapache2-mod-php libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64 libevent-extra-2.1-7t64 libltdl7 liblua5.4-0 libmodbus5 libmysqlclient21 libodbc2
  libopenipmi0t64 libsnmp-base libsnmp40t64 php-bcmath php-common php-curl php-gd php-ldap
  php-mbstring php-mysql php-xml php8.3-bcmath php8.3-cli php8.3-common php8.3-curl php8.3-gd
  php8.3-ldap php8.3-mbstring php8.3-mysql php8.3-opcache php8.3-readline php8.3-xml snmpd ssl-cert
  zabbix-agent zabbix-apache-conf zabbix-frontend-php zabbix-server-mysql zabbix-sql-scripts
0 upgraded, 48 newly installed, 0 to remove and 127 not upgraded.
Need to get 31.6 MB of archives.
After this operation, 151 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9,116 B]
Get:5 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.5 [1,329 kB]
Get:7 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.5 [163 kB]
Get:8 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.5 [97.1 kB]
Get:9 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.5 [90.2 kB]
Get:10 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libsnmp-base all 5.9.4+dfsg-1.1ubuntu3 [206 kB]
Get:11 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libsnmp40t64 amd64 5.9.4+dfsg-1.1ubuntu3 [1,066 kB]
Get:12 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 snmpd amd64 5.9.4+dfsg-1.1ubuntu3 [59.6 kB]
Get:13 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libevent-extra-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [64.2 kB]
Get:14 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmysqlclient21 amd64 8.0.40-0ubuntu0.24.04.1 [1,254 kB]
Get:15 https://repo.zabbix.com/zabbix/7.2/stable/ubuntu noble/main amd64 zabbix-server-mysql amd64 1:7.2.2-1ubuntu24.04 [1,754 kB]
Get:16 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:17 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libodbc2 amd64 2.3.12-1ubuntu0.24.04.1 [158 kB]
Get:18 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 libopenipmi0t64 amd64 2.0.33-1.1build3 [502 kB]
Get:19 http://ca.archive.ubuntu.com/ubuntu noble/universe amd64 fping amd64 5.1-1 [32.0 kB]
Get:20 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 fonts-dejavu-extra all 2.37-8 [1,947 kB]
Get:21 http://ca.archive.ubuntu.com/ubuntu noble/universe amd64 fonts-dejavu all 2.37-8 [3,020 B]
Get:22 http://ca.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:23 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.3 [739 kB]
Get:24 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.3 [372 kB]
```

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Zabbix Database Installation

1. Good work, next step would be to create the initial zabbix database on the ubuntu server VM.

Make sure you have database server up and running from step 3 above.

First switch to the root user using sudo su.

```
$ sudo su
```

```
fuadm@fuad:~$ sudo su
[sudo] password for fuadm:
root@fuad:/home/fuadm#
```

2. Then use the command mysql (we don't need to add -p because we do not have a password configured)

```
# mysql
```

```
root@fuad:/home/fuadm# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11096
Server version: 8.0.32-0ubuntu0.20.04.2 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

SAMPLE SCREENSHOT

3. Next step would be to create our database using the command

```
>create database zabbix character set utf8mb4 collate utf8mb4_bin;
```

```
mysql> create database zabbix character set utf8mb4 collate utf8mb4_bin;
```

SAMPLE SCREENSHOT

4. Next would be to create our zabbix user using the command below.

```
>create user zabbix@localhost identified by 'zabbix';
```

```
mysql> create user zabbix@localhost identified by 'zabbix';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
```

**Password I
used is zabbix**

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5. Next step would be grant our zabbix user the privileges to access the database using the command below.

```
> grant all privileges on zabbix.* to zabbix@localhost;
```

```
mysql> grant all privileges on zabbix.* to zabbix@localhost;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>
```

6. Next, set the trust functions using the following command below.

```
> set global log_bin_trust_function_creators = 1;
```

```
mysql> set global log_bin_trust_function_creators = 1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> quit  
Bye  
root@fuad:/home/fuadm#
```

SAMPLE SCREENSHOT

7. On Zabbix server host import initial schema and data. You will be prompted to enter your newly created password from step 4 above. In my own case, I used zabbix as my password.

First change your directory using the command

```
# cd /usr/share/zabbix-sql-scripts/mysql/
```

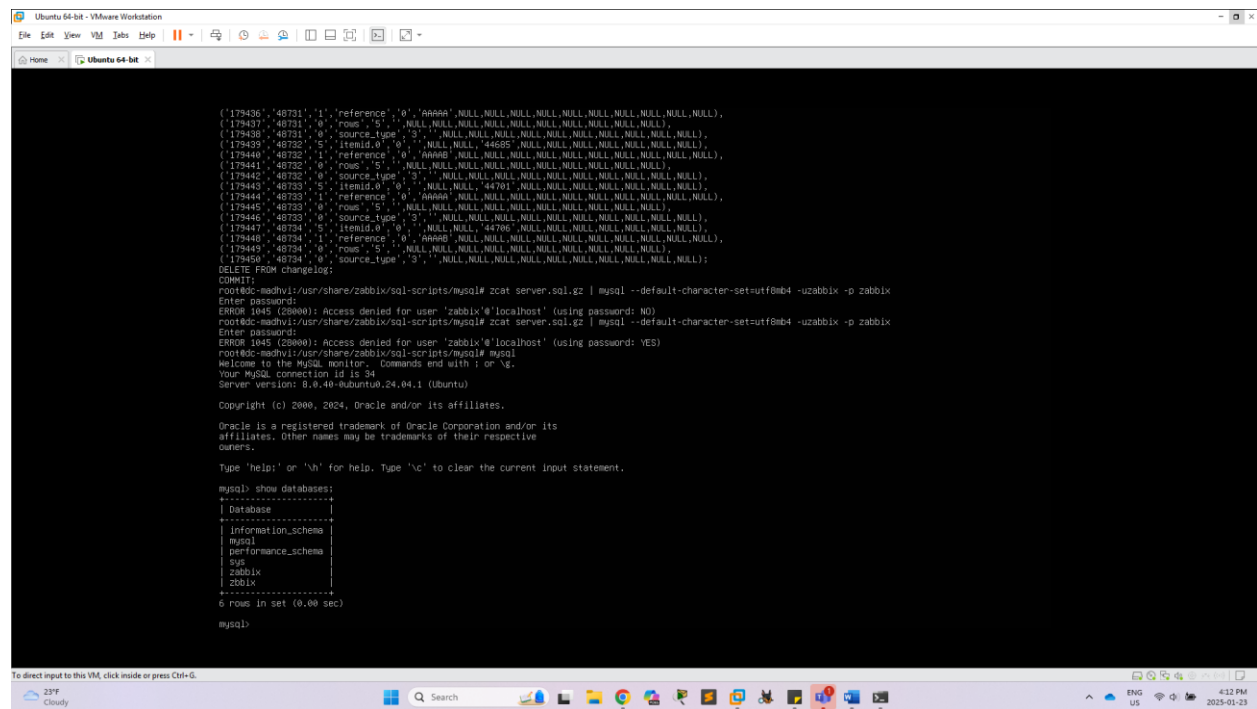
Then run the command import initial schema and data

```
# zcat server.sql.gz | mysql Zabbix
```

```
root@fuad:/home/fuadm# cd /usr/share/zabbix-sql-scripts/mysql/  
root@fuad:/usr/share/zabbix-sql-scripts/mysql# ls  
double.sql history_pk_prepare.sql proxy.sql server.sql.gz  
root@fuad:/usr/share/zabbix-sql-scripts/mysql# zcat server.sql.gz | mysql zabbix
```

**This may take some time so
be patient**

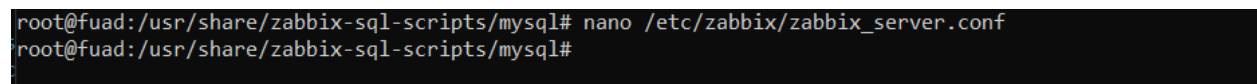
[Attach Your Screenshot]



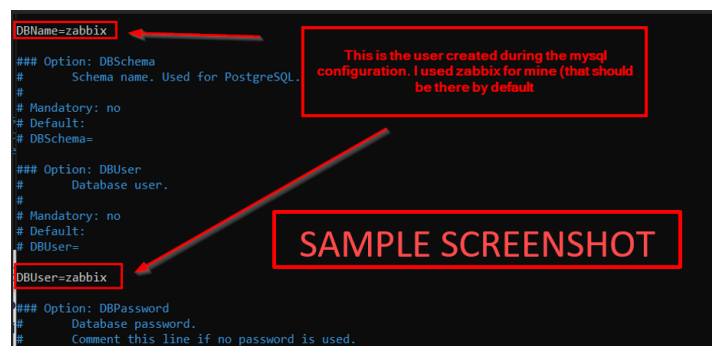
- Next step would be to configure the database for the Zabbix server

Open the Zabbix server configuration file to confirm the dbuser and password. Use the below command

```
# nano /etc/zabbix/zabbix_server.conf
```



Confirming the name used for the mysql database

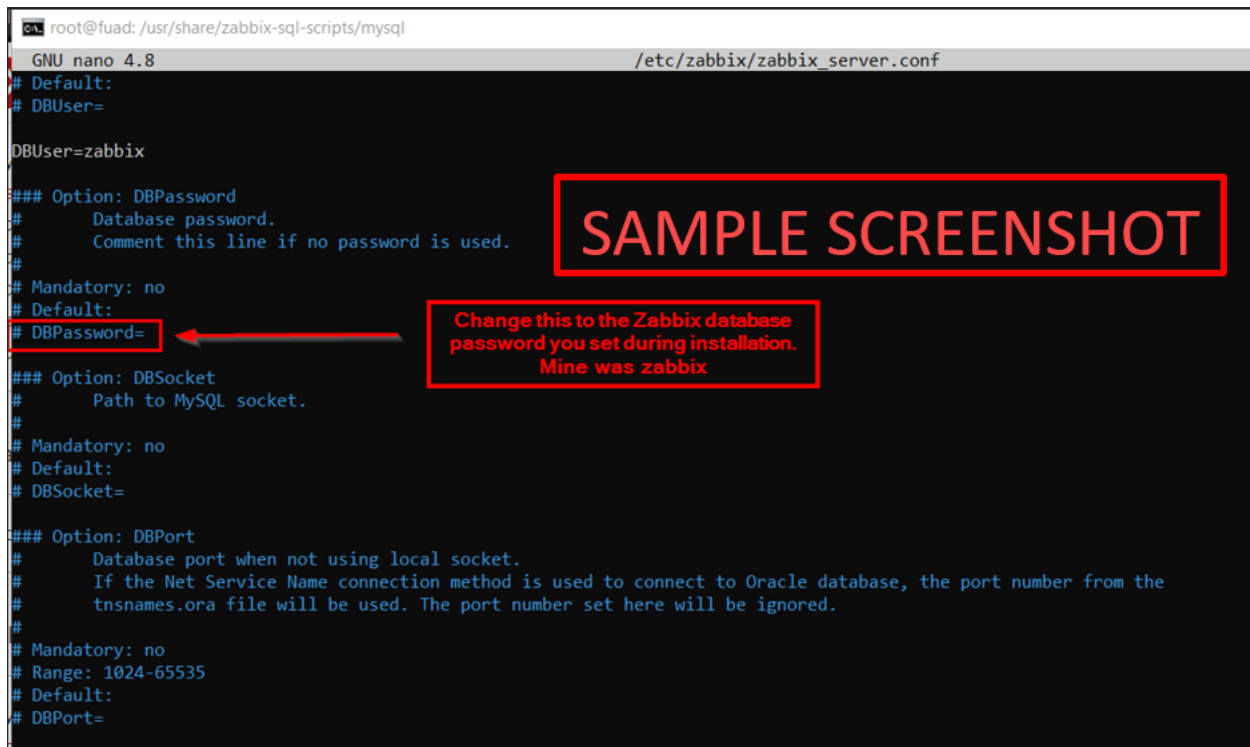


Before update

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```
root@fuad: /usr/share/zabbix-sql-scripts/mysql
GNU nano 4.8 /etc/zabbix/zabbix_server.conf
# Default:
# DBUser=

DBUser=zabbix

### Option: DBPassword
# Database password.
# Comment this line if no password is used.
#
# Mandatory: no
# Default:
# DBPassword=
### Option: DBSocket
# Path to MySQL socket.
#
# Mandatory: no
# Default:
# DBSocket=

### Option: DBPort
# Database port when not using local socket.
# If the Net Service Name connection method is used to connect to Oracle database, the port number from the
# tnsnames.ora file will be used. The port number set here will be ignored.
#
# Mandatory: no
# Range: 1024-65535
# Default:
# DBPort=
```

After update



```
root@fuad: /usr/share/zabbix-sql-scripts/mysql
GNU nano 4.8 /etc/zabbix/zabbix_server.conf
# Default:
# DBUser=

DBUser=zabbix

### Option: DBPassword
# Database password.
# Comment this line if no password is used.
#
# Mandatory: no
# Default:
# DBPassword=zabbix
### Option: DBSocket
# Path to MySQL socket.
#
# Mandatory: no
# Default:
# DBSocket=
```

9. Save the configuration file using `ctrl o` on your keyboard and use `ctrl x` to exit, then start the Zabbix server and agent processes and check the status using the command below. The status should show **active (running)**.

Command to restart the Zabbix server, Zabbix agent and apache2

```
# systemctl restart zabbix-server zabbix-agent apache2
```

Command to enable the Zabbix server, Zabbix agent and the apache2

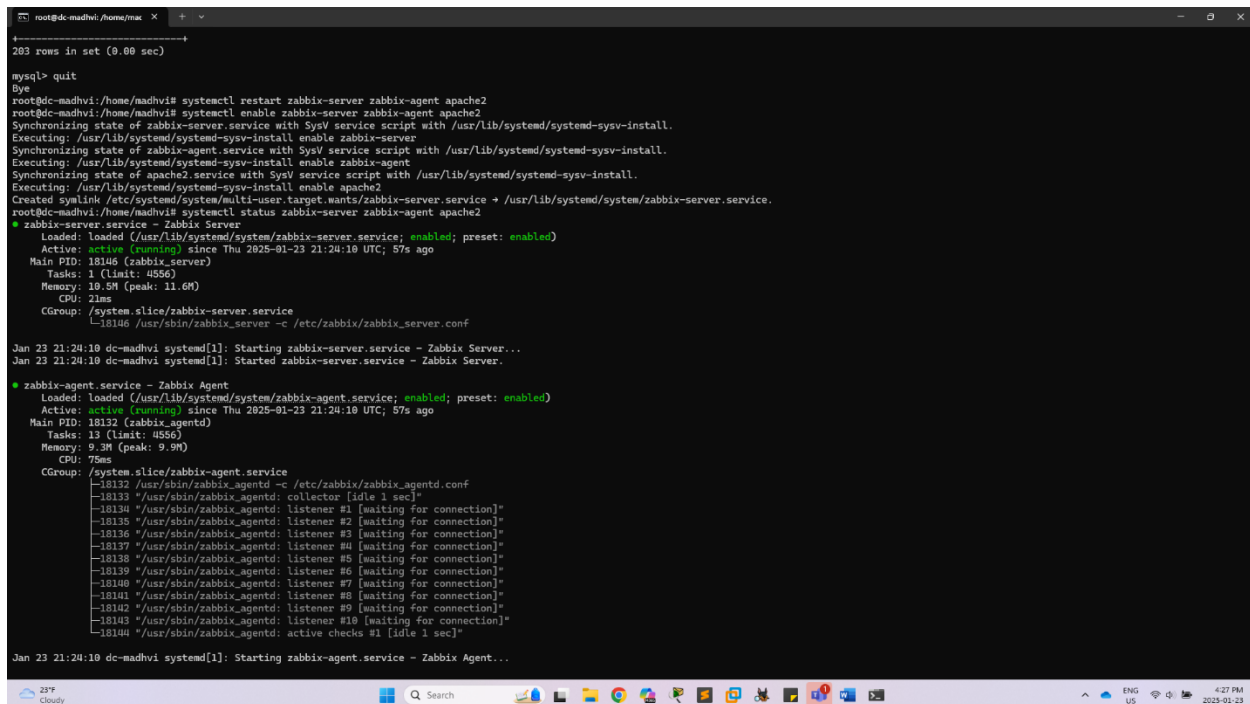
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```
# systemctl enable zabbix-server zabbix-agent apache2
```

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```
root@dc-madhvi:/home/madhvi# systemctl restart zabbix-server zabbix-agent apache2
root@dc-madhvi:/home/madhvi# systemctl enable zabbix-server zabbix-agent apache2
Executing: /usr/lib/systemd/systemd-sysv-install enable zabbix-server
Synchronizing state of zabbix-agent.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable zabbix-agent
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-server.service → /usr/lib/systemd/system/zabbix-server.service.
root@dc-madhvi:/home/madhvi# systemctl status zabbix-server zabbix-agent apache2
● zabbix-server.service - Zabbix Server
   Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-01-23 21:24:10 UTC; 57s ago
     Main PID: 18146 (zabbix_server)
       Tasks: 1 (Limit: 4556)
      Memory: 10.5M (peak: 11.6M)
         CPU: 21ms
    CGroup: /system.slice/zabbix-server.service
            └─18146 /usr/sbin/zabbix_server -c /etc/zabbix/zabbix_server.conf

Jan 23 21:24:10 dc-madhvi systemd[1]: Starting zabbix-server.service - Zabbix Server...
Jan 23 21:24:10 dc-madhvi systemd[1]: Started zabbix-server.service - Zabbix Server.

● zabbix-agent.service - Zabbix Agent
   Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-01-23 21:24:10 UTC; 57s ago
     Main PID: 18132 (zabbix_agentd)
       Tasks: 13 (Limit: 4556)
      Memory: 9.3M (peak: 9.9M)
         CPU: 75ms
    CGroup: /system.slice/zabbix-agent.service
            └─18132 /usr/sbin/zabbix_agentd -c /etc/zabbix/zabbix_agentd.conf
                  └─18133 "/usr/sbin/zabbix_agentd: collector [idle 1 sec]"
                  └─18134 "/usr/sbin/zabbix_agentd: listener #1 [waiting for connection]"
                  └─18135 "/usr/sbin/zabbix_agentd: listener #2 [waiting for connection]"
                  └─18136 "/usr/sbin/zabbix_agentd: listener #3 [waiting for connection]"
                  └─18137 "/usr/sbin/zabbix_agentd: listener #4 [waiting for connection]"
                  └─18138 "/usr/sbin/zabbix_agentd: listener #5 [waiting for connection]"
                  └─18139 "/usr/sbin/zabbix_agentd: listener #6 [waiting for connection]"
                  └─18140 "/usr/sbin/zabbix_agentd: listener #7 [waiting for connection]"
                  └─18141 "/usr/sbin/zabbix_agentd: listener #8 [waiting for connection]"
                  └─18142 "/usr/sbin/zabbix_agentd: listener #9 [waiting for connection]"
                  └─18143 "/usr/sbin/zabbix_agentd: listener #10 [waiting for connection]"
                  └─18144 "/usr/sbin/zabbix_agentd: active checks #1 [idle 1 sec]"

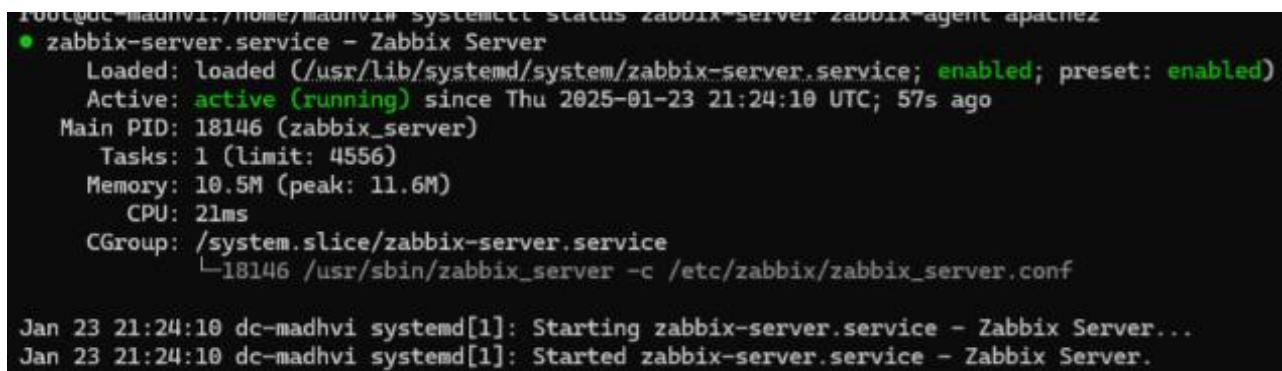
Jan 23 21:24:10 dc-madhvi systemd[1]: Starting zabbix-agent.service - Zabbix Agent...
```

Command to check the status of your zabbix server, zabbix agent and apache2. Note use the down button on your keyboard to see the other agents running

```
# systemctl status zabbix-server zabbix-agent apache2
```

Zabbix server running

[Attach Your Screenshot]



```
root@dc-madhvi:/home/madhvi# systemctl status zabbix-server zabbix-agent apache2
● zabbix-server.service - Zabbix Server
   Loaded: loaded (/usr/lib/systemd/system/zabbix-server.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-01-23 21:24:10 UTC; 57s ago
     Main PID: 18146 (zabbix_server)
       Tasks: 1 (Limit: 4556)
      Memory: 10.5M (peak: 11.6M)
         CPU: 21ms
    CGroup: /system.slice/zabbix-server.service
            └─18146 /usr/sbin/zabbix_server -c /etc/zabbix/zabbix_server.conf

Jan 23 21:24:10 dc-madhvi systemd[1]: Starting zabbix-server.service - Zabbix Server...
Jan 23 21:24:10 dc-madhvi systemd[1]: Started zabbix-server.service - Zabbix Server.
```

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Zabbix agent running

[Attach Your Screenshot]

```
● zabbix-agent.service - Zabbix Agent
   Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-01-23 21:24:10 UTC; 57s ago
     Main PID: 18132 (zabbix_agentd)
        Tasks: 13 (limit: 4556)
       Memory: 9.3M (peak: 9.9M)
          CPU: 75ms
     CGroup: /system.slice/zabbix-agent.service
             └─18132 /usr/sbin/zabbix_agentd -c /etc/zabbix/zabbix_agentd.conf
             └─18133 "/usr/sbin/zabbix_agentd: collector [idle 1 sec]"
             └─18134 "/usr/sbin/zabbix_agentd: listener #1 [waiting for connection]"
             └─18135 "/usr/sbin/zabbix_agentd: listener #2 [waiting for connection]"
             └─18136 "/usr/sbin/zabbix_agentd: listener #3 [waiting for connection]"
             └─18137 "/usr/sbin/zabbix_agentd: listener #4 [waiting for connection]"
             └─18138 "/usr/sbin/zabbix_agentd: listener #5 [waiting for connection]"
             └─18139 "/usr/sbin/zabbix_agentd: listener #6 [waiting for connection]"
             └─18140 "/usr/sbin/zabbix_agentd: listener #7 [waiting for connection]"
             └─18141 "/usr/sbin/zabbix_agentd: listener #8 [waiting for connection]"
             └─18142 "/usr/sbin/zabbix_agentd: listener #9 [waiting for connection]"
             └─18143 "/usr/sbin/zabbix_agentd: listener #10 [waiting for connection]"
             └─18144 "/usr/sbin/zabbix_agentd: active checks #1 [idle 1 sec]"
```

Apache2 service running

[Attach Your Screenshot]

```
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-01-23 21:24:10 UTC; 57s ago
     Docs: https://httpd.apache.org/docs/2.4/
     Main PID: 18154 (apache2)
        Tasks: 6 (limit: 4556)
       Memory: 13.7M (peak: 13.9M)
          CPU: 37ms
     CGroup: /system.slice/apache2.service
             └─18154 /usr/sbin/apache2 -k start
             └─18156 /usr/sbin/apache2 -k start
             └─18157 /usr/sbin/apache2 -k start
```

lines 1-50

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Access to the Zabbix web interface

1. Now go to your browser on your host machine to access the Zabbix setup using the IP address of your Virtual machine. **(Note: On your host machine e.g. windows operating system)**

Go to your zabbix VM to run the ip a command to check what your zabbix server IP address is. Mine is 192.168.80.206. Yours would be different.

<http://yourZabbixIP/zabbix>

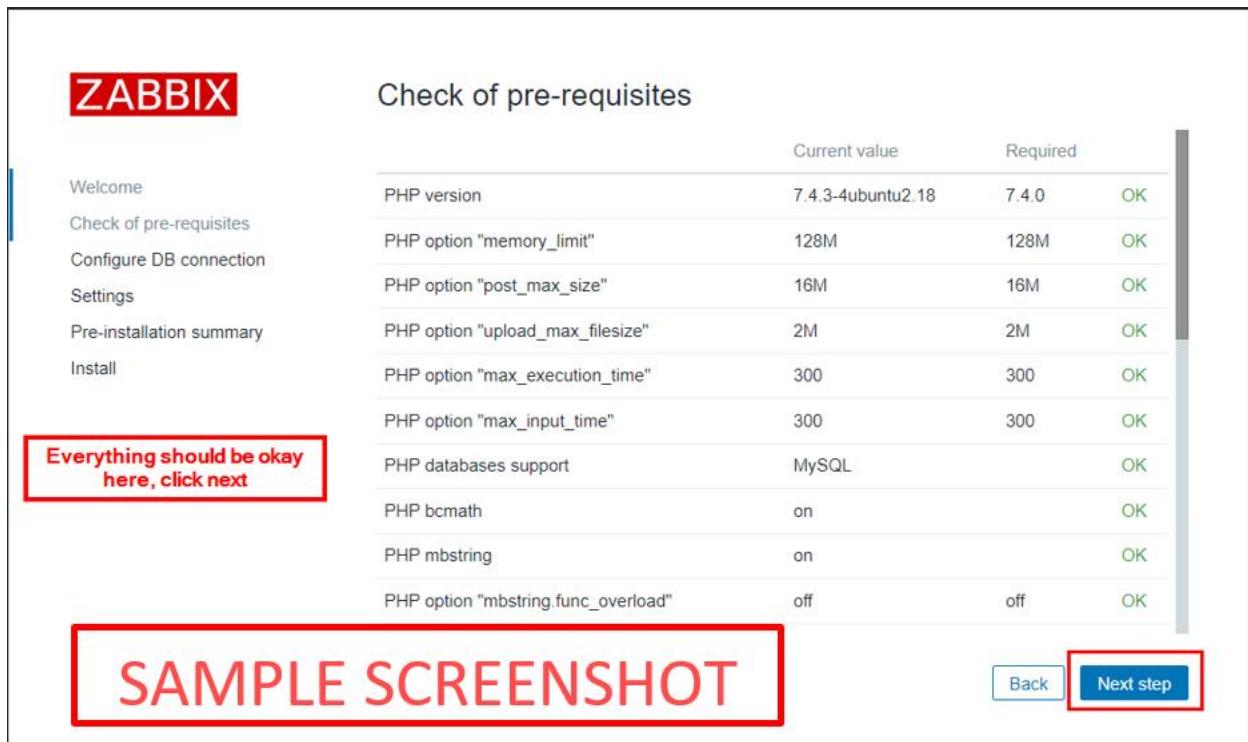
<http://192.168.80.206/zabbix> **(Note, your IP address would be different from mine and ensure you add the /zabbix to it)**

2. Once the setup page comes up, you can leave the default language as English and click on next.



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3. All pre-requisites should be showing OK, then click next.



ZABBIX Check of pre-requisites

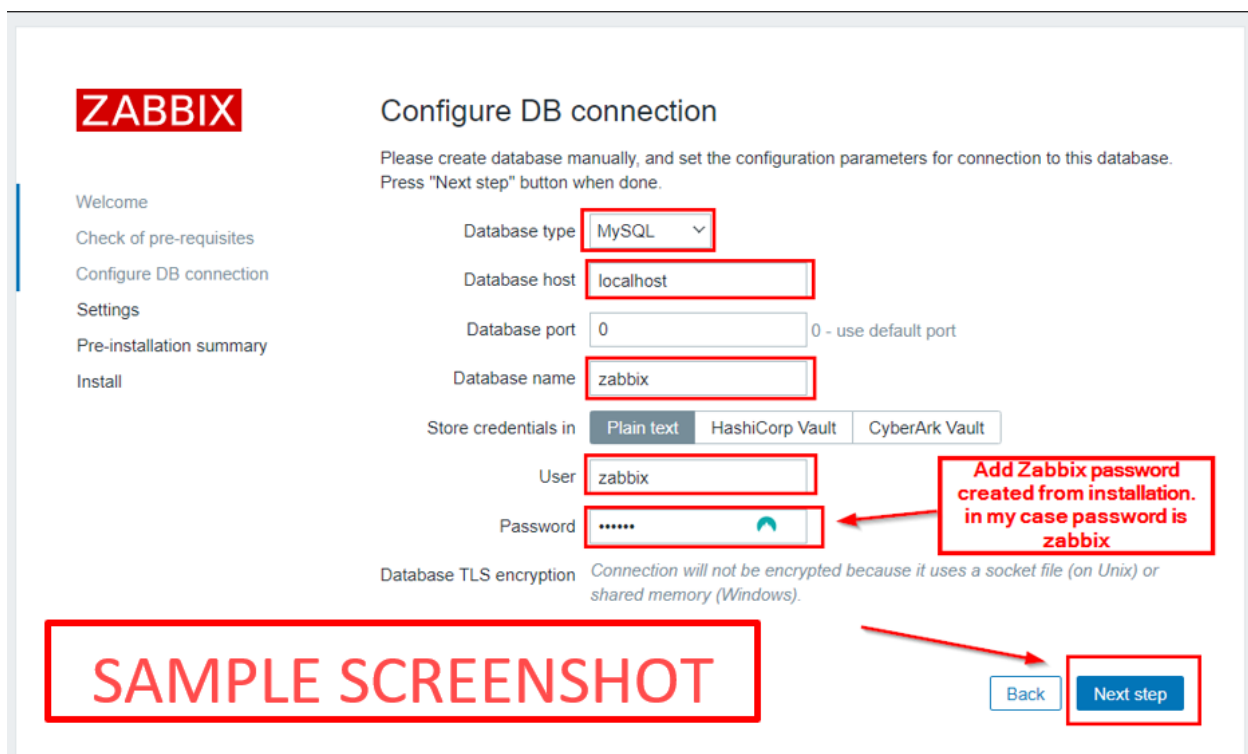
Everything should be okay here, click next

	Current value	Required	
PHP version	7.4.3-4ubuntu2.18	7.4.0	OK
PHP option "memory_limit"	128M	128M	OK
PHP option "post_max_size"	16M	16M	OK
PHP option "upload_max_filesize"	2M	2M	OK
PHP option "max_execution_time"	300	300	OK
PHP option "max_input_time"	300	300	OK
PHP databases support	MySQL		OK
PHP bcmath	on		OK
PHP mbstring	on		OK
PHP option "mbstring.func_overload"	off	off	OK

SAMPLE SCREENSHOT

Back Next step

4. Ensure every box is checked correctly and add the zabbix user and password created from the initial installation. Click next if everything is complete



ZABBIX Configure DB connection

Please create database manually, and set the configuration parameters for connection to this database. Press "Next step" button when done.

Database type: MySQL

Database host: localhost

Database port: 0 - use default port

Database name: zabbix

Store credentials in: Plain text HashiCorp Vault CyberArk Vault

User: zabbix

Password:

Database TLS encryption: Connection will not be encrypted because it uses a socket file (on Unix) or shared memory (Windows).

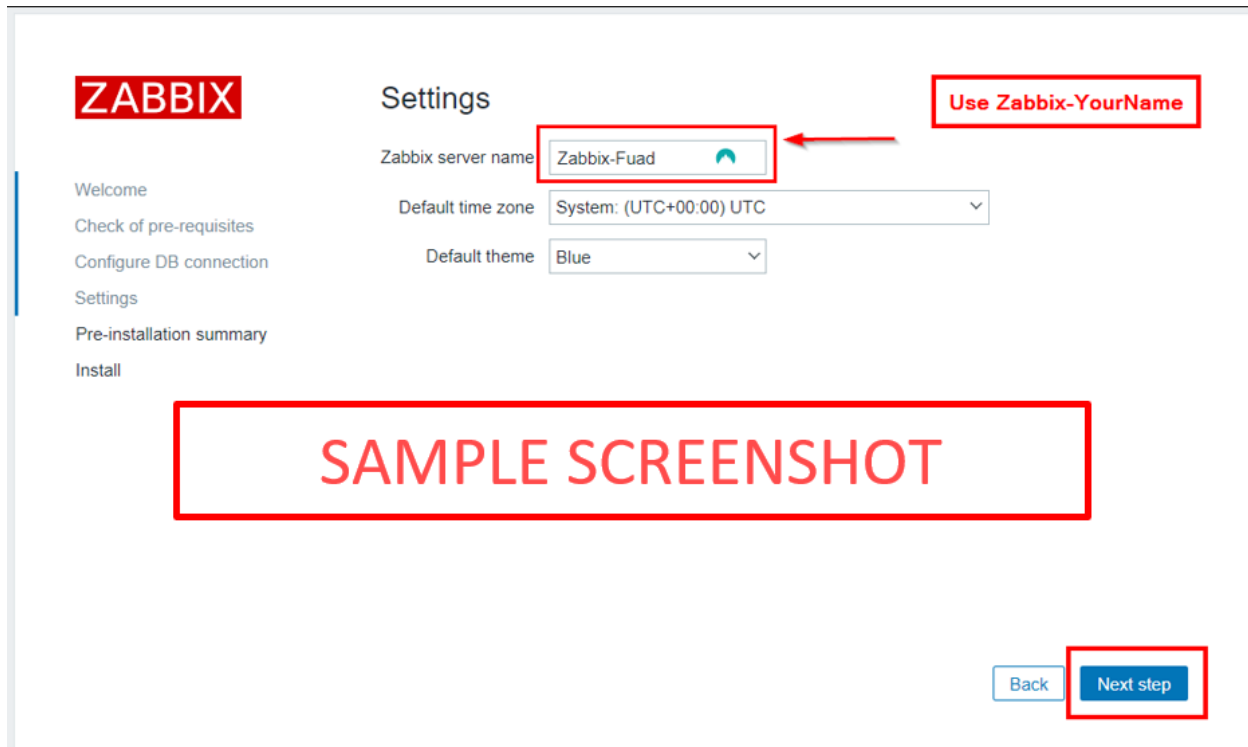
Add Zabbix password created from installation. in my case password is zabbix

SAMPLE SCREENSHOT

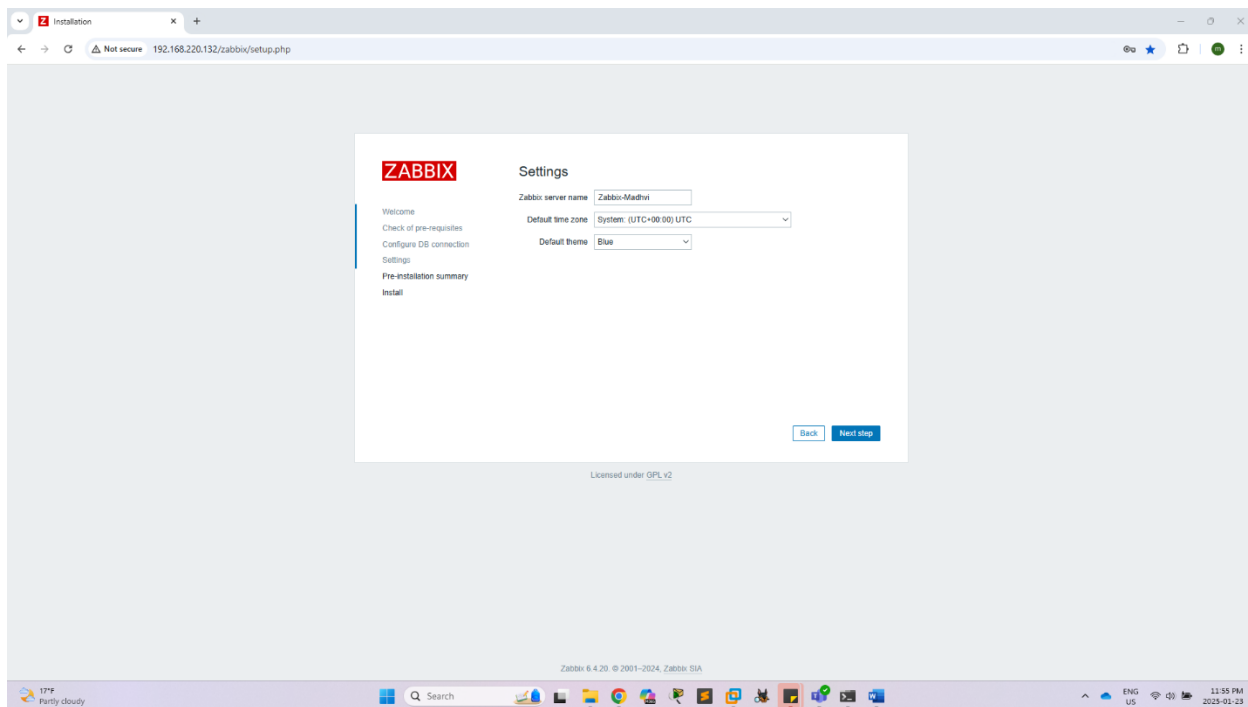
Back Next step

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5. Make your Zabbix server name your name with this naming convention “Zabbix-Yourname” for the Zabbix server name. In my case I used Zabbix-Fuad. Then lick next



[Attach Your Screenshot]

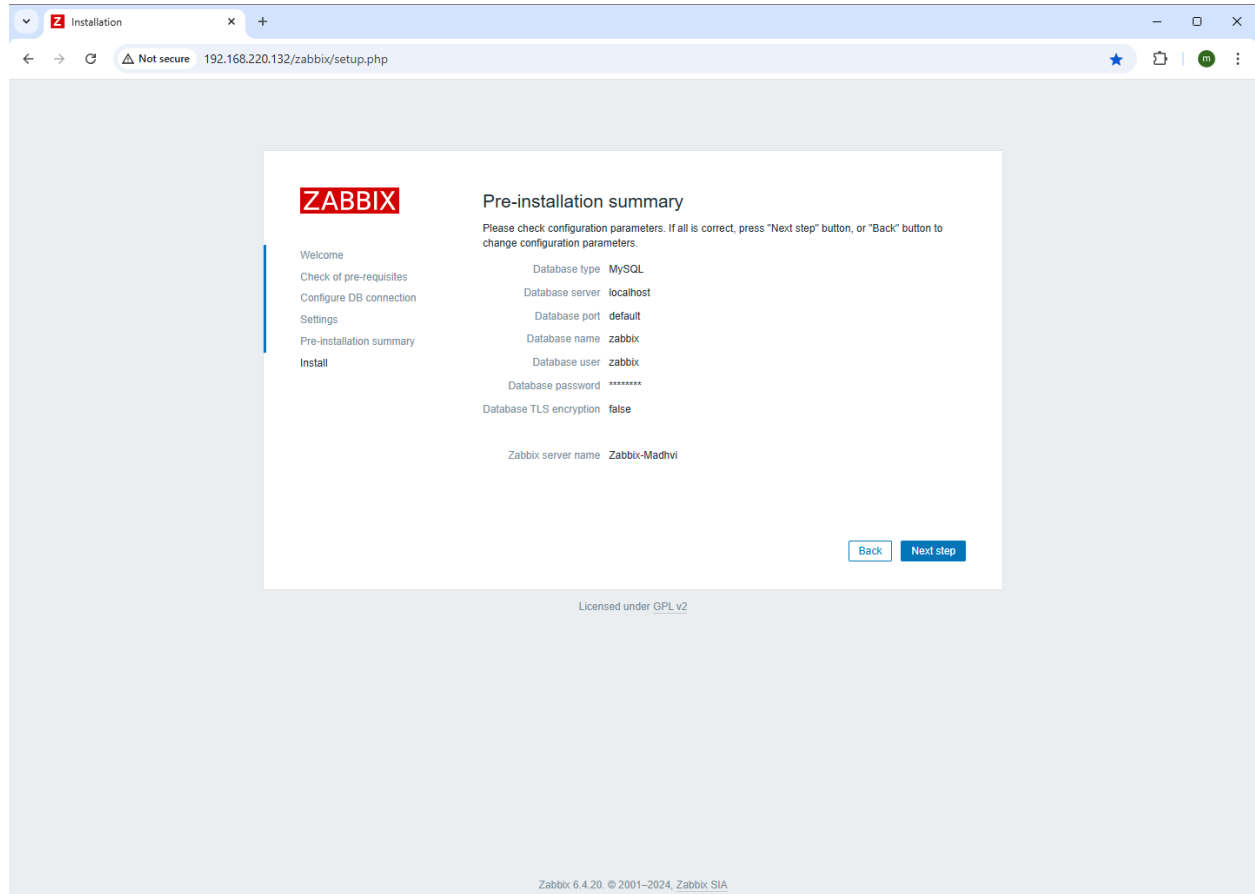


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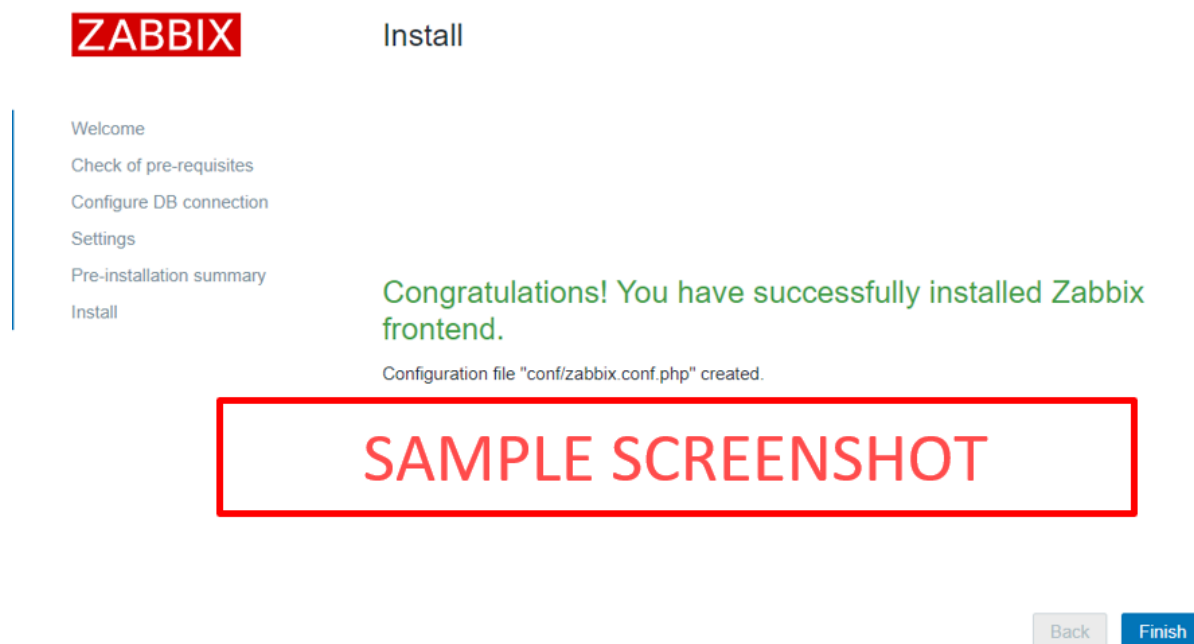
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6. Check the pre-installation summary to make sure there are no errors.

[Attach Your Screenshot]



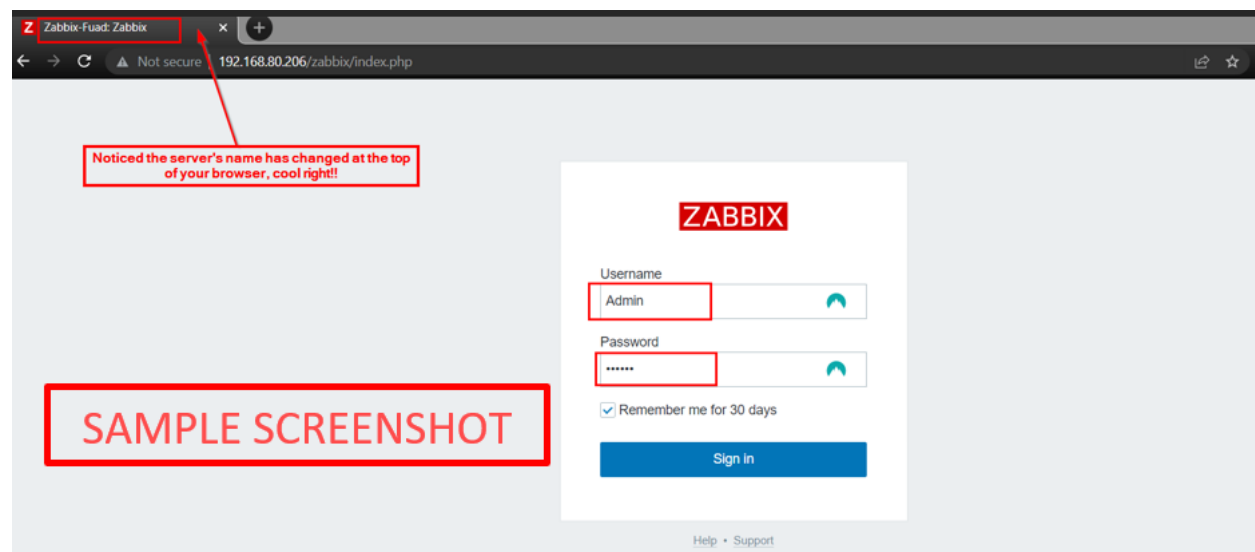
7. Then you should see the congratulations! message of a complete installation of Zabbix. Click finish



8. You will notice your server's name has changed to the name you added. Pretty cool right!! Input the default zabbix username and password to login.

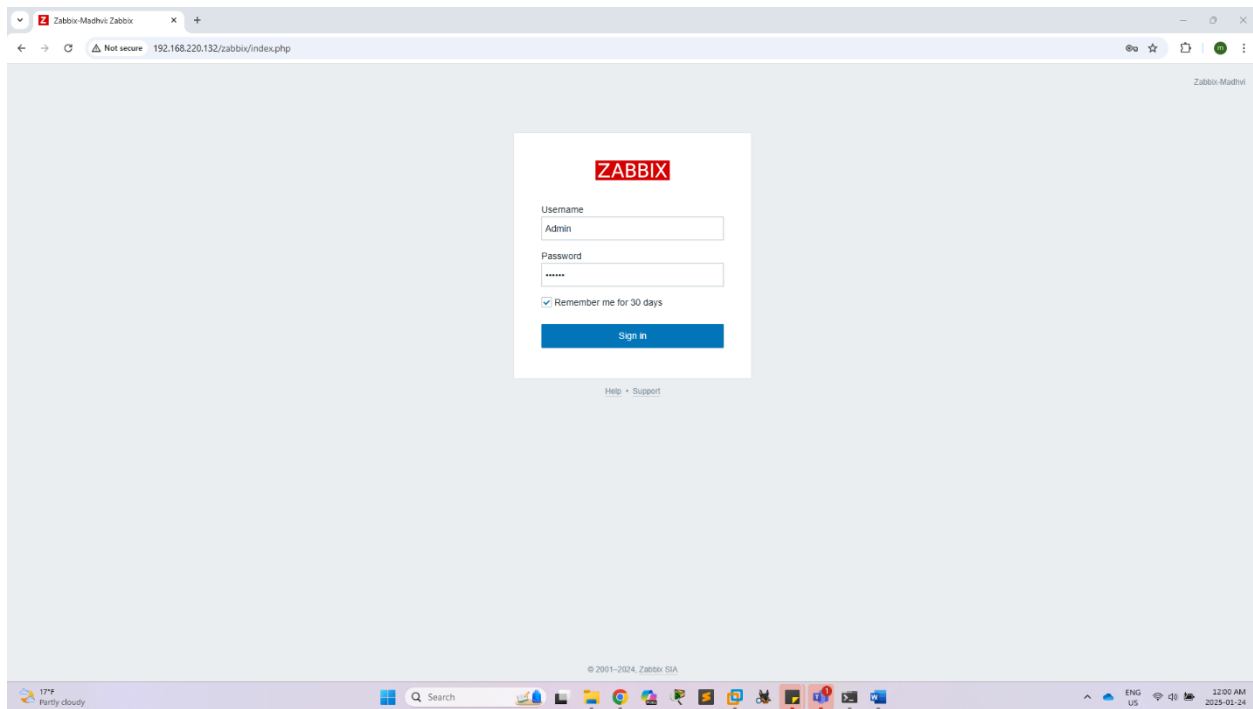
Username: Admin

Password: zabbix

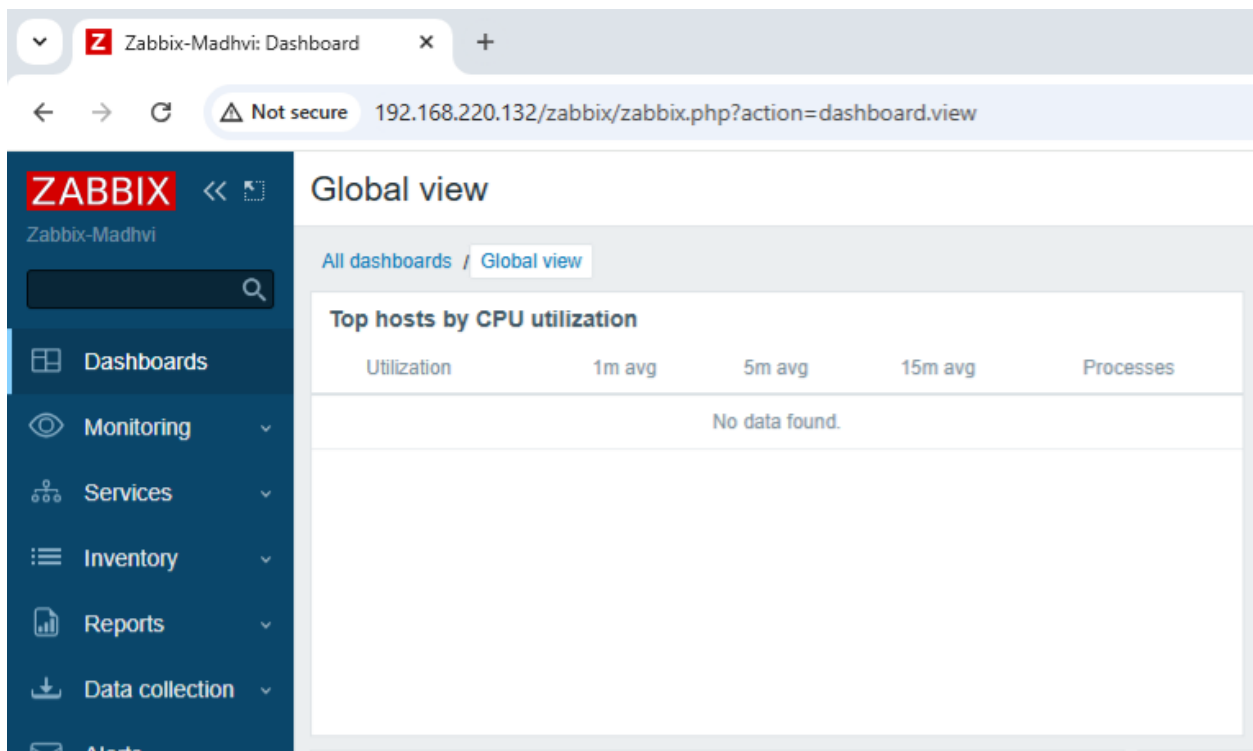


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[Attach Your Screenshot]



[Attach Your Screenshot] **Note: Screenshot must show the Zabbix dashboard and your name on the top right side of the dashboard.**



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Congratulations, you have Zabbix installed

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 10 Screenshots (10 marks each)
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professor's from lab manual. They are for your reference only and would be considered an academic integrity breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**

Lab 2: Install Zabbix SNMP Agent on a Windows Server 2019

Lab Workbook

Lab 2 – Madhvi_100997946

Name – Student ID	Course Code	Weight
Click or tap here to enter text.	INFT1202	5%

Part 1: Windows 2019 Server Installation

Part 2: Heavy loader application installation on Windows server

Part 3: Enable Zabbix monitoring on Windows 2019 server

Part 4: Execute the load runner on Windows to generate load

Lab Objectives

- Install Windows 2019 Virtual Machine (VM).
- Monitor the Windows server 2019 server using Zabbix
- Run Heavy loader application on Windows server 2019

Lab Materials

- Please name your Lab workbook titled “Lab 2 - <name> <Student ID>”.
 - Please include all relevant screenshots on Lab workbook for submission (provided on dc connect)
- Install Windows Server 2019 VM
- Use Zabbix to monitor the Windows Server 2019

Lab Requirements

- VMware workstation 16 pro
- Windows server 2019
- Zabbix installation from Lab 1
- Heavy loader application

Lab 2 Week 3 – Zabbix SNMP Agent Installation

PART 1: WINDOWS 2019 SERVER INSTALLATION

1. Go to your Azure cloud portal “<https://portal.azure.com/>” to download windows server 2019 **(If you have a previously installed Windows 2019/2022 server, you can clone that for this task)**
2. Download windows server 2019/2022.

Software

Education



Windows Server 2019 Datacenter (updated Sept 2019)

Windows Server 2019 Standard is ideal for customers with low density or non-virtualized environments. Windows Server 2019 offers additional features in Standard and Datacenter editions. Features exclusive to the Windows Server 2016 Datacenter edition include Shielded Virtual Machines, software-defined networking, Storage Spaces Direct, and Storage Replica. While no features from the Windows Server 2016 R2 Standard edition have been removed, we have added features like Nano Server and unlimited Windows Server containers to the Windows Server 2019 Standard edition.

Operating System

Windows

Product language

English

System

64 bit

View Key

3. Install the windows 2019 server on your VMware Workstation (I believe you all should know how to install this by now)

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Microsoft Azure Education | Software

Software	Category	Operating System	Architecture
Machine Learning Server for Windows	AI + Machine Learning	Windows	64 bit
Microsoft Hyper-V Server 2019 (updated Sept 2019)	Compute	Windows	64 bit
Microsoft R Server 9.1.0 for Hadoop	Database	Windows	64 bit
Microsoft R Server 9.1.0 for Linux	Database	Linux	64 bit
Microsoft R Server 9.1.0 for Teradata	Database	Windows	64 bit
Microsoft R Server 9.1.0 for Windows	Database	Windows	64 bit
SharePoint Server Subscription Edition	Productivity Tools	Windows	64 bit
SharePoint Server Subscription Edition	Productivity Tools	Windows	64 bit
Skype for Business Server 2019	Productivity Tools	Windows	64 bit
Windows Server 2022 Datacenter	Operating System	Windows	64 bit
Windows Server 2019 Datacenter (updated Sept 2019)	Operating System	Windows	64 bit
Windows Server 2019 Essentials (updated Sept 2019)	Operating System	Windows	64 bit
Windows Server 2019 Features on Demand	Operating System	Windows	64 bit
Windows Server 2019 Language Pack	Operating System	Windows	64 bit
Windows Server 2022 Standard	Operating System	Windows	64 bit
Windows Server 2019 Standard (updated Sept 2019)	Operating System	Windows	64 bit
SQL Server 2019 Standard	Database	Windows	64 bit

Software
Education

Windows Server 2019 Datacenter (updated Sept 2019)

Windows Server 2019 Standard is ideal for customers with low density or non-virtualized environments. Windows Server 2019 offers additional features in Standard and Datacenter editions. Features exclusive to the Windows Server 2016 Datacenter edition include Shielded Virtual Machines, software-defined networking, Storage Spaces Direct, and Storage Replica. While no features from the Windows Server 2016 R2 Standard edition have been removed, we have added features like Nano Server and unlimited Windows Server containers to the Windows Server 2019 Standard edition.

Operating System
Windows

Product language
English

System
64 bit

[View Key](#)

[Help improve this page](#)

4. After the installation, Go to VM > Settings of the new installed Windows server virtual machine to change the memory and processor settings.
 - Memory: 4 GB
 - Processors: 1

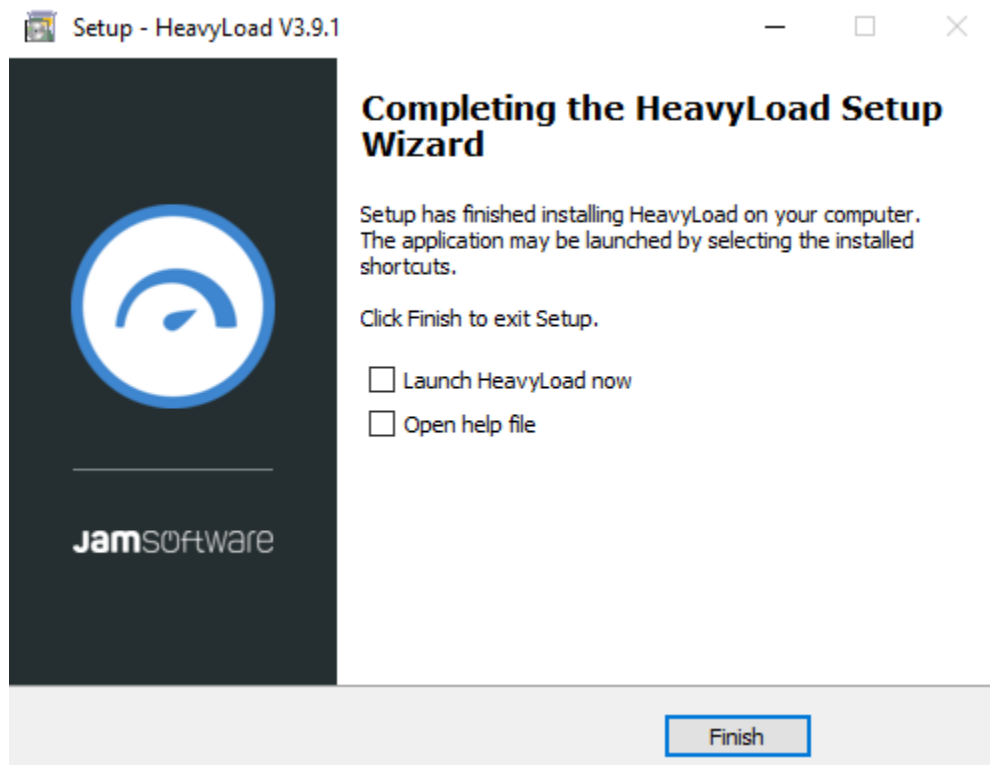
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PART 2: HEAVY LOADER APPLICATION INSTALLATION ON WINDOWS SERVER

Turn on your newly installed Windows server to install the heavy loader application.

1. Go to the website <https://www.jam-software.com/heavyload?ca=1> to download the heavy loader application setup.
2. This can be downloaded on your desktop or download folder. After download, double click on the setup to install.
3. Click yes to allow the application to run and follow the prompt
4. You can choose to create a desktop shortcut after installation.
5. Uncheck the Launch heavy Load now and click finish

[Attach Screenshot]



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PART 3: ENABLE ZABBIX MONITORING ON WINDOWS 2019 SERVER

Go to the windows server 2019 and install SNMP using the server manager

1. Install SNMP Service in Windows Server 2019.
 - Open Server Manager
 - Click Add roles and features
 - Under Before you begin click Next
 - Under Select installation type click Next
 - Under Select destination server click Next
 - Under Select server roles click Next
 - Under Select features select SNMP Service and click Add Features and then click Next
2. Select SNMP Service
 - Under Confirm installation selection click, Install
 - Wait until the installation is finished
3. Once the installation is done, Open the Services applet (click on the Start menu and search for services) and click Apply and OK to start the 'SNMP Service'
4. Click the 'Agent' tab and specify the following details:
 - Contact: ZabbixMonitor
 - Location: Windows-SERVER
 - Here, select the following services:
 - Physical, Applications, Internet, End-to-end, then click 'OK'.

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- Click the 'Security' tab on the same dialog box and click Add, to add the community.
- Input the community name is 'ZabbixMonitor' here and the Community rights: 'READ ONLY' then click 'Add'.
- Select 'Accept SNMP packets from any host'
- The next step would be to add your Windows server Machine as a host in your Zabbix instance.

Go to the 'Data Collection' tab in your Zabbix, then click 'Hosts'.

- Click 'Create host'.
- Use the host name: Windows_SERVER_Yourname, in my own case, I used Windows_SERVER_Fuad.

[Attach Screenshot]

New host

Host IPMI Tags Macros Inventory Encryption Value mapping

* Host name Windows_SERVER_Madhvi

Visible name Windows_SERVER_Madhvi

Templates type here to search Select

* Host groups type here to search Select

Interfaces No interfaces are defined.

Add

Description

Monitored by proxy (no proxy)

Enabled ☒

Add Cancel

- Then click on select by the templates section
- Click 'Select' to add the Template group as 'Templates/Operating Systems'.
- Choose 'Windows SNMP' and click 'Select'
- Now click the 'Select' in *Host groups.
- In Host groups, choose everything and click 'Select'.
- In Interfaces, click 'Add' to configure the SNMP Interface.
- Select SNMP from the drop down.
- Configure the following values for SNMP Interface:

- IP address: (Your Windows Server 2019 IP Address) mine was 192.168.108.132
- Port: 161
- SNMP version: SNMPv2
- SNMP community: {\$SNMP_COMMUNITY}

Go to 'Macros' tab then enter the following details:

- Macro: {\$SNMP_COMMUNITY}

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- Value: ZabbixMonitor

After Editing

[Attach Screenshot]

The screenshot shows the 'New host' configuration page in Zabbix. The 'Host name' is 'Windows_SERVER_Madhvi'. The 'Visible name' is also 'Windows_SERVER_Madhvi'. The 'Templates' section shows 'Windows by SNMP' selected. The 'Host groups' section shows 'Applications', 'Databases', 'Discovered hosts', 'Hypervisors', 'Linux servers', 'Virtual machines', and 'Zabbix servers' selected. The 'Interfaces' section shows 'SNMP' selected with IP address '192.168.220.135', DNS name, 'Connect to' set to 'IP', 'Port' set to '161', and 'Default' set to 'Remove'. The 'SNMP version' is 'SNMPv2' and the 'SNMP community' is '{\$SNMP_COMMUNITY}'. The 'Max repetition count' is '10' and 'Use combined requests' is checked. The 'Description' field is empty. The 'Monitored by proxy' is set to '(no proxy)' and 'Enabled' is checked. The 'Add' button is visible at the bottom right.

Then click 'Add'.

19. Confirm that the status of your newly added agent is enabled.

[Attach Screenshot]

The screenshot shows the 'Hosts' page in Zabbix. A green banner at the top indicates 'Host added'. The 'Hosts' table lists the following hosts:

Name	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption	Info	Tags
Windows_SERVER_Madhvi	15	7	1	3	Web	192.168.220.135:161		Windows by SNMP	Enabled	SNMP	None		
Zabbix server	136	74	27	5	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX	None		

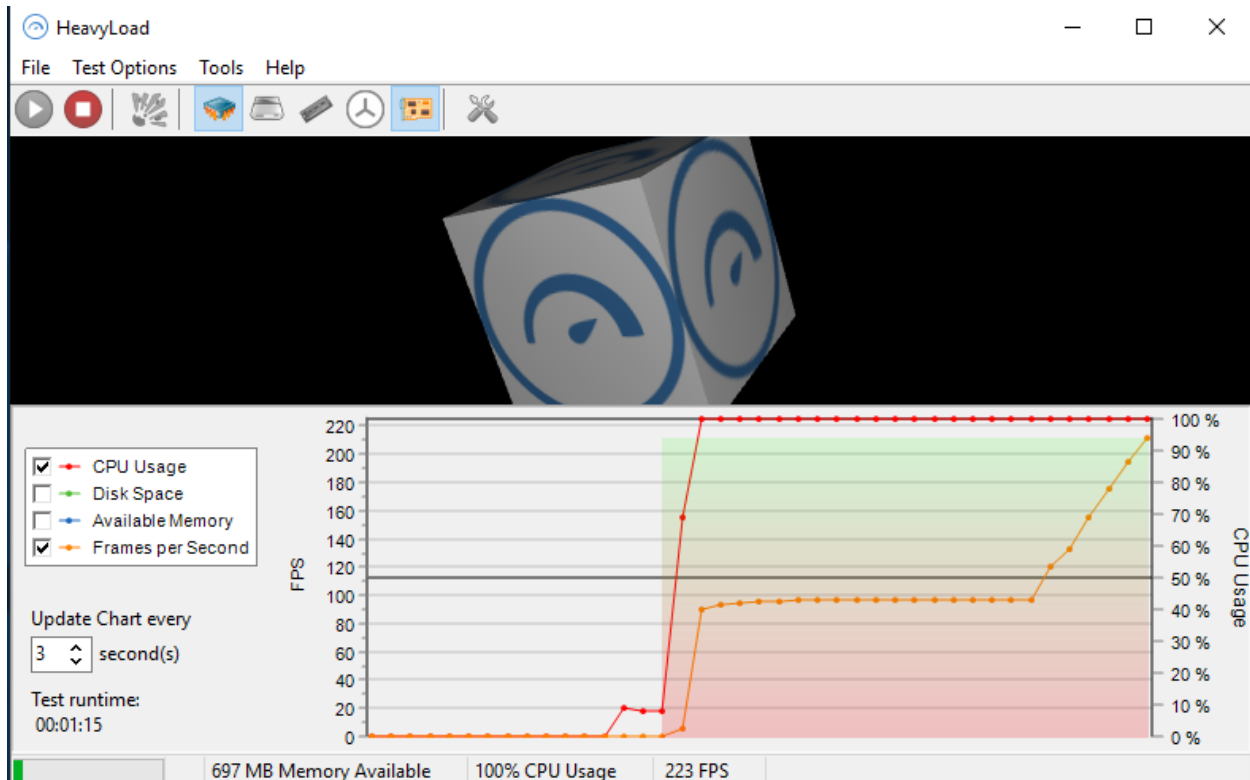
The table shows that the newly added host 'Windows_SERVER_Madhvi' is in the 'Enabled' status. The 'Zabbix server' is also listed with a status of 'Enabled'.

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PART 4: EXECUTE THE LOAD RUNNER ON WINDOWS TO GENERATE LOAD

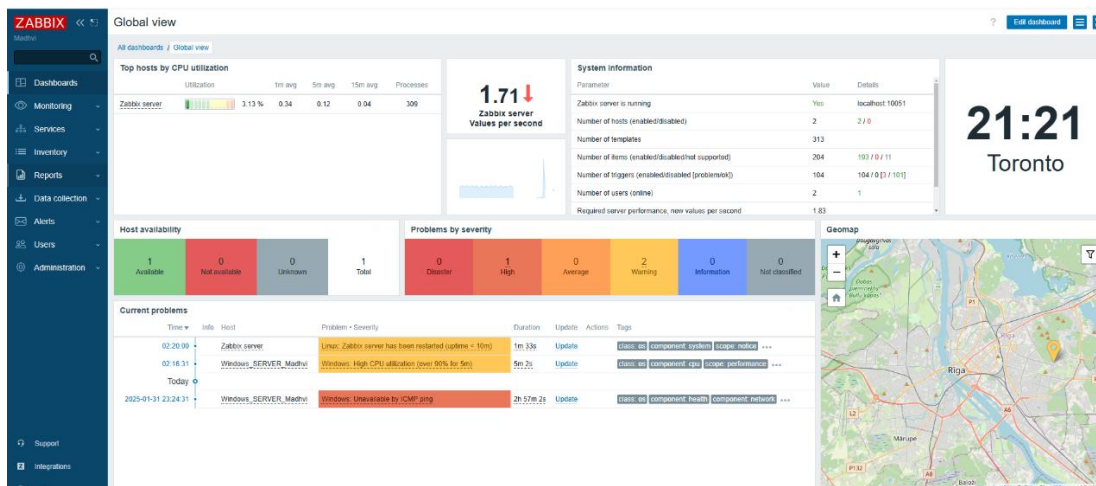
1. Launch the LoadRunner application from your windows server machine to generate a high CP usage.
2. This would simulate generating load on the Windows Machine and check the status in Zabbix.
3. Else manually run some applications on Windows-SERVER to generate load.

[Attach Screenshot]



4. Go over to your Zabbix instance through your browser and click on Dashboard, you should see a current problems tab. Showing a high memory utilization has been reported on your Zabbix server.

[Attach Screenshot]

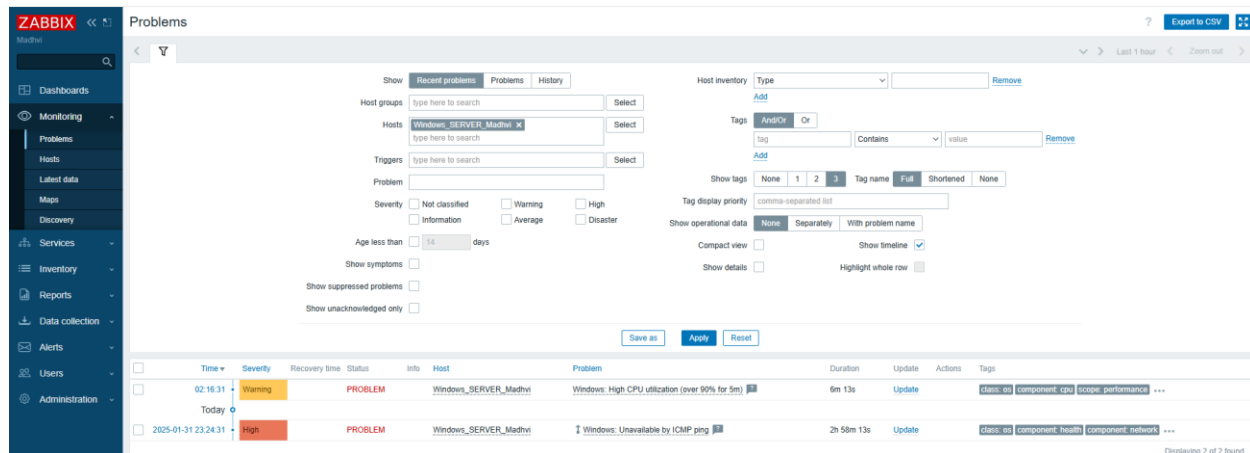


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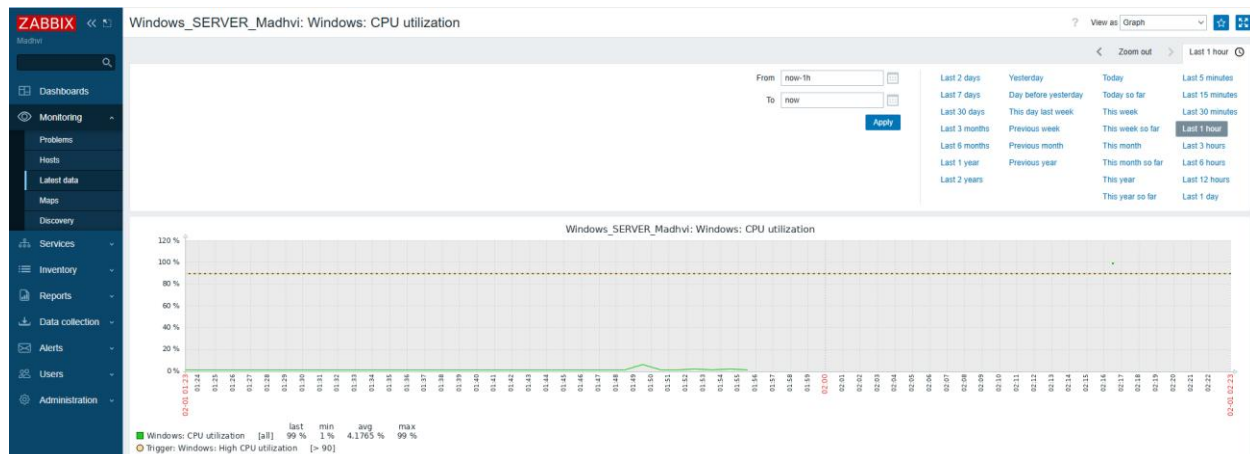
- Click on the **Monitoring > hosts** in the left menu tab section to see the problems row. Click on the problems to see the details of the problem reported.
- Check to see the Status of the problem

[Attach Screenshot]



- Click on the problem section to click the History > Physical Memory: Memory Utilization.
- This would display a graph of the high CPU utilization usage of your windows server 2019.

[Attach Screenshot]



- Go over to your windows 2019 server to stop the heavy loader application
- Once you have stopped the heavy loader software, go back to your Zabbix instance to see the status of your problems. You should see a RESOLVED status and PROBLEM status.

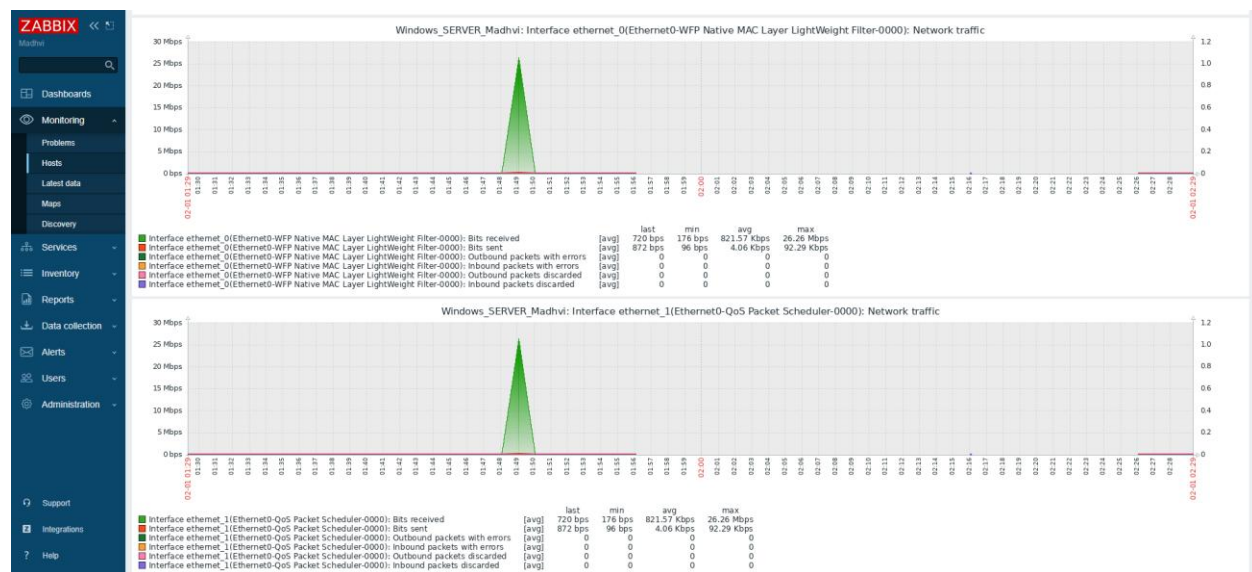
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[Attach Screenshot]

The screenshot shows the Zabbix 'Problems' page. The left sidebar contains navigation links: Dashboards, Monitoring, Problems (selected), Hosts, Latest data, Maps, Discovery, Services, Inventory, Reports, Data collection, Alerts, Users, and Administration. The main content area has a search bar and filters for Host groups, Hosts (Windows_SERVER_Madnvi), Triggers, and Severity (Warning, High, Information, Average, Disaster). It also includes filters for Age less than (14 days), Show symptoms, Show suppressed problems, and Show unacknowledged only. On the right, there are filters for Host inventory, Tags, Show tags, Tag display priority, Show operational data, Compact view, and Show details. Below the filters is a table of problems with columns: Time, Severity, Recovery time, Status, Info, Host, Problem, Duration, Update, Actions, and Tags. The table shows one problem: 'Windows_SERVER_Madnvi: Windows: High CPU utilization (over 90% for 5m)' with a severity of 'Warning', recovery time of '02:25:23', and status 'RESOLVED'.

11. Click on the hosts from the left menu tab. Navigate to the graph row, to open the graphs.
12. The page containing all the graphs from your windows 2019 server should be displayed

[Attach Screenshot]



Congratulations, you have completed the Zabbix monitoring configuration on your Windows Server 2019.

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Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- **Marks distribution:**
 - a) 10 Screenshots (100 marks)
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professor's from lab manual. They are for your reference only and would be considered an academic integrity breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**